

BSc Thesis Proposal

# **SETU: A Selective Triage-and-Correct Framework for Hallucination Mitigation in Bengali Small Language Models**

*Uncertainty-Calibrated, Type-Aware Correction across Native Bengali and Bangla-English Code-  
Mixed Question Answering, Summarization, and Reasoning*

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*SETU (সেতু, “bridge”) = Selective Evidence-grounded Triage with Uncertainty-aware correction*

## 0. Reference Verification Note (read first)

**The proposal is built only on papers whose venue, authors, and claims were verified.** An earlier topic-suggestion PDF that circulated in the group contained unreliable citations (placeholder arXiv IDs such as “2403.xxxxx” and mismatched author/venue attributions). None of those unverified entries are used here. Every base and supporting paper below was cross-checked against the uploaded PDFs or the original venue page (arXiv, ACL Anthology, IEEE Xplore, OpenReview) in July 2026.

## 1. Introduction and Motivation

Large Language Models (LLMs) and their smaller variants (SLMs, roughly 1–3B parameters) frequently produce fluent but factually wrong or ungrounded text — a failure called **hallucination**. A medical query bot that invents a drug dose, a summarizer that adds facts absent from the source, or a math tutor that silently corrupts one arithmetic step can all cause real harm. Two large 2026 studies frame the problem for our setting precisely: (i) smaller models hallucinate significantly more than larger ones, and (ii) hallucination is worse and harder to measure in low-resource, non-English languages, where uncertainty signals are often mis-calibrated.

Bengali is the sixth most spoken language in the world, yet until mid-2026 no framework even **measured** hallucination in Bengali LLMs. BenHalluEval (2026) closed that measurement gap for four tasks (Generative QA, Bangla-English code-mixed QA, summarization, reasoning) and showed two uncomfortable facts: hallucination calibration varies wildly across models, and chain-of-thought prompting — the usual cheap fix — helped in only 9 of 21 model-task combinations and hurt in 12.

**The open problem this thesis attacks:** Bengali has an evaluation benchmark but **no mitigation framework**. BenHalluEval explicitly lists “consistency-based mitigation strategies” as future work. The only Bengali self-correction work (BanglaForge, 2025) is restricted to code generation, and notes that Bengali generation systems still lack built-in grounding, correction, or iteration mechanisms. Reducing — not just detecting — hallucination in Bengali SLMs is unexplored territory.

## 2. Why This Is Method-Level Novelty (not a dataset swap)

A common weak thesis pattern is: take an existing English method, run it on a Bengali dataset, report a higher accuracy. That is replication, not contribution. SETU is different in **mechanism**, and the contribution survives even if datasets change:

- **A type-aware triage router.** Every existing mitigation method applies a single uniform fix — either retrieval, or self-verification. SETU first classifies each hallucination-prone claim as data-driven (missing knowledge) or reasoning-driven (logical error), then routes it to the correct repair. No prior Bengali or SLM system does this routing.
- **A first calibration study of uncertainty signals for Bengali SLMs.** Multilingual studies report that uncertainty is mis-calibrated in low-resource languages, but no one has measured which signal (semantic entropy, self-consistency, verbalized confidence) is actually reliable in Bengali. This is a dataset-independent scientific finding.

- **Mitigation, not detection.** The entire Bengali hallucination literature to date is detection/evaluation only. SETU produces corrected or abstained answers and is scored on how much it lowers hallucination.
- **Abstention as a first-class output.** When neither repair resolves a flagged claim, SETU makes the model say “আমি নিশ্চিত ন” (I am not sure) instead of committing a confident error — aligned with the 2026 shift from chasing zero hallucination to managing uncertainty.
- **Code-mixed mitigation.** Bangla-English code-mixed text is heavily used online and is a distinct task in BenHalluEval, but mitigation on code-mixed input is completely unexamined.

**Proof of contribution:** an ablation study turns each SETU component off in isolation, so the improvement is demonstrably in the *mechanism*, not in data selection.

### 3. Research Questions

1. **RQ1.** Which uncertainty signals are calibrated enough to flag hallucination-prone claims in Bengali SLMs, and does calibration differ between native Bengali and code-mixed input?
2. **RQ2.** Can hallucinations be reliably triaged as data-driven vs reasoning-driven in a black-box SLM setting without model-internal access?
3. **RQ3.** Does type-aware routing (retrieval for data-driven, belief-consistency for reasoning-driven) reduce hallucination more than a single uniform fix, measured by dual-track BenHalluScore?
4. **RQ4.** Does uncertainty-grounded abstention improve overall reliability without excessively refusing answerable questions?

## 4. Base Papers (verified, 2024–2026)

SETU stands on six primary papers. Each contributes one reusable component; SETU's novelty is the combination and the routing between them.

### 4.1 BenHalluEval — arXiv 2605.31483 (updated Jun 2026) [evaluation protocol]

- **What it gives us:** the dual-track evaluation protocol (Track A on ground-truth answers to catch false alarms; Track B on hallucinated candidates to catch misses) and the BenHalluScore metric =  $\frac{1}{2} \times (\text{Track A error} + \text{Track B error})$ . Also the four Bengali tasks and 12,000 hallucinated candidates.
- **Gap we exploit:** detection/evaluation only; CoT mitigation is unreliable (9/21 improve, 12/21 degrade); mitigation is left as future work.

### 4.2 HalluGuard — ICLR 2026 [triage inspiration]

- **What it gives us:** the formal decomposition of hallucination into a data-driven term (flawed/missing knowledge) and a reasoning-driven term (inference-time instability). SETU borrows this taxonomy as the basis of its triage router.
- **Gap we exploit:** HalluGuard is a detection score requiring NTK/internal access on English benchmarks; SETU makes the data-vs-reasoning decision black-box and lightweight, then acts on it to correct output.

### 4.3 BTPROP (Belief Tree Propagation) — NAACL 2025 [reasoning-driven fix]

- **What it gives us:** a consistency-based verification idea — expand a claim into logically related statements and use NLI + confidence consistency to judge truthfulness. SETU uses a lightweight version as the reasoning-driven repair.
- **Gap we exploit:** English-only, and its full belief tree is compute-heavy; SETU applies a shallow, SLM-friendly consistency check triggered only for reasoning-flagged claims.

### 4.4 HalluSearch — SemEval-2025 Task 3 (Mu-SHROOM) [data-driven fix]

- **What it gives us:** atomic factual-claim splitting + retrieval-augmented verification, with fallback strategies when search coverage is thin. SETU adopts claim splitting and retrieval verification for data-driven claims.
- **Gap we exploit:** it reports degraded performance in low-online-coverage languages; SETU adds a cross-lingual (Bengali→English) retrieval fallback to compensate for the sparse Bengali corpus.

### 4.5 “How Much Do LLMs Hallucinate across Languages?” (MFAVA) — arXiv 2502.12769 (updated Feb 2026) [motivation]

- **What it gives us:** empirical grounding — smaller models hallucinate more; broader-language models hallucinate more; hallucination rate does not simply track a language's data size; and hallucination-rate estimation must correct for detector precision/recall. Justifies focusing on SLMs and on careful metric design.

#### 4.6 Cost-Efficient Cross-Lingual RAG for Bengali — arXiv 2601.02065 (Jan 2026) [cross-lingual grounding]

- **What it gives us:** evidence that translating Bengali queries to English, retrieving from a high-resource corpus, and returning in Bengali improves low-resource grounding. SETU turns this static pipeline into an adaptive fallback used only when the Bengali corpus is insufficient.

## 5. Positioning: How SETU Differs from Every Base Paper

| Work                             | Bengali | Mitigation<br>(not just<br>detection) | Type-aware<br>triage (data<br>vs reasoning) | Uncertainty<br>calibration<br>study | Code-mixed | Abstention |
|----------------------------------|---------|---------------------------------------|---------------------------------------------|-------------------------------------|------------|------------|
| BenHalluEval (2026)              | Yes     | No                                    | No                                          | No                                  | Yes (eval) | No         |
| HalluGuard (ICLR 2026)           | No      | No                                    | Yes<br>(detection)                          | No                                  | No         | No         |
| BTPROP (NAACL 2025)              | No      | No                                    | No                                          | Partial                             | No         | No         |
| HalluSearch (SemEval 2025)       | No      | Detection                             | No                                          | No                                  | No         | No         |
| Cross-Lingual Bengali RAG (2026) | Yes     | Yes                                   | No                                          | No                                  | No         | No         |
| SETU (this thesis)               | Yes     | Yes                                   | Yes                                         | Yes                                 | Yes        | Yes        |

Table 1: SETU is the only framework that combines Bengali + mitigation + type-aware triage + uncertainty calibration + code-mixed coverage + abstention.

**One-sentence claim for the defense:** “To the best of our knowledge, SETU is the first framework to reduce (not merely detect) hallucination in Bengali small language models, using a novel uncertainty-triggered triage that routes each flagged claim to type-specific correction — cross-lingual retrieval for knowledge gaps, belief-consistency for reasoning errors — and abstains when unresolved, evaluated with dual-track BenHalluScore across native and code-mixed Bengali.”

## 6. Complete Reading List (team split)

Suggested split for a three-member team: each member owns one group plus shares the two Bengali base papers.

### Group A — Base papers (read fully)

1. BenHalluEval: A Multi-Task Hallucination Evaluation Framework for LLMs on Bengali (arXiv 2605.31483, 2026).
2. HalluGuard: Demystifying Data-Driven and Reasoning-Driven Hallucinations in LLMs (ICLR 2026).
3. A Probabilistic Framework for LLM Hallucination Detection via Belief Tree Propagation — BTPROP (NAACL 2025).
4. HalluSearch: A Search-Enhanced RAG Pipeline for Hallucination Detection (SemEval-2025 Task 3).
5. How Much Do LLMs Hallucinate across Languages? — MFAVA (arXiv 2502.12769, 2026).
6. Cost-Efficient Cross-Lingual RAG for Low-Resource Languages: A Bengali Case Study (arXiv 2601.02065, 2026).

### Group B — Methods & signals (read methods sections)

1. MALTO at SemEval-2024 Task 6 (SHROOM): synthetic-data + NLI ensemble for hallucination detection.

2. SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection (Manakul et al., EMNLP 2023) — self-consistency signal.
3. Detecting Hallucinations in LLMs Using Semantic Entropy (Farquhar et al., Nature 2024) — uncertainty signal.
4. Chain-of-Verification Reduces Hallucination (Dhuliawala et al., ACL 2024) — verify-then-revise baseline.

### **Group C — Bengali & multilingual context (read + cite in Ch.2)**

1. Generation, Analysis, and Detection of LLM-Manipulated Bangla News (Akther et al., IEEE Access, Dec 2025) — knowledge-based detection (NER + retrieval + consistency).
2. Social Bias in LLMs for Bangla (Sadhu et al., LoResLM 2025) — first Bangla LLM social-behaviour study; motivates reliability work.
3. MultiHaluDet: Multilingual Hallucination Detection via LLM Hidden-State Probing (MeLLM 2026).
4. BanglaMedQA / BanglaMMedBench: RAG strategies for Bangla biomedical QA (arXiv 2511.04560, 2025).
5. BanglaBERT (Bhattacharjee et al., NAACL 2022) and TyDiQA-GoldP (Clark et al., 2020) — model and data references.

## 7. Methodology — The SETU Pipeline

### 7.1 High-level flow

Bengali (or code-mixed) query → SLM draft answer → (1) atomic claim decomposition → (2) per-claim uncertainty scoring → flag high-uncertainty claims → (3) triage each flagged claim as data-driven or reasoning-driven → (4) targeted correction: cross-lingual retrieval for data-driven, belief-consistency check for reasoning-driven → (5) if unresolved, abstain on that claim → reassemble a corrected, grounded answer.

### 7.2 Stage-by-stage

1. **Draft generation.** A Bengali-capable SLM (candidates: Qwen2.5-1.5B/3B-Instruct, Gemma-2-2B, and a Bangla-centric model in the TigerLLM line) produces an initial answer at 4-bit precision.
2. **Atomic claim decomposition (Bengali-adapted).** Split the answer into single-fact claims (following the HalluSearch / FActScore idea), so each claim can be verified independently. Prompt templates are authored and validated by native Bengali speakers on the team.
3. **Per-claim uncertainty scoring (RQ1).** Compute three signals per claim — semantic entropy (multiple samples clustered by NLI), self-consistency (agreement across samples), and verbalized confidence — and measure their calibration (Expected Calibration Error, AUROC vs gold) separately for native and code-mixed input. This calibration study is a standalone contribution.
4. **Triage / routing (RQ2).** A lightweight classifier labels each flagged claim data-driven (answer needs external facts — entity/date/quantity) or reasoning-driven (a logical/arithmetic step is wrong). Signals: retrieval-answerability probe, claim type (factual vs inferential), and self-contradiction under paraphrase. Grounded in HalluGuard’s decomposition but done black-box.
5. **Data-driven correction.** Retrieve from a Bengali Wikipedia FAISS index (embeddings: BGE-M3 / multilingual-e5, plus BM25). If Bengali retrieval score is below threshold, fall back to cross-lingual retrieval: translate the claim to English, retrieve from English Wikipedia, return grounding in Bengali (adaptive version of the 2026 Bengali cross-lingual RAG pipeline). Revise the claim to match retrieved evidence.
6. **Reasoning-driven correction.** Run a shallow belief-consistency check (lightweight BTPROP): generate a few logically related sub-claims, use an NLI model to test entailment/contradiction, and revise the claim if it is inconsistent with its own supported premises.
7. **Abstention (RQ4).** If a flagged claim remains unresolved after correction, replace it with an explicit “আমি নিশ্চিত না” marker rather than emitting a confident guess.
8. **Reassembly.** Recombine verified, corrected, and abstained claims into the final Bengali answer.

### 7.3 Datasets (all public, from the base papers)

| Task          | Dataset                                | Role                                                              |
|---------------|----------------------------------------|-------------------------------------------------------------------|
| Generative QA | TyDiQA-GoldP (Bengali)                 | Question–passage–answer triples; primary QA evaluation.           |
| Code-Mixed QA | Code-mixed TyDiQA (BenHalluEval style) | Bangla–English Roman-script questions; unique mitigation setting. |
| Summarization | BanglaCHQ-Summ                         | Medical query–summary pairs; faithfulness evaluation.             |

| Task             | Dataset                           | Role                                                 |
|------------------|-----------------------------------|------------------------------------------------------|
| Reasoning        | SOMADHAN                          | Bengali math word problems with chain-of-thought.    |
| Detection val.   | BenHalluEval 12K candidates       | If released: validate uncertainty/triage components. |
| Retrieval corpus | Bengali + English Wikipedia dumps | Chunked to passages, FAISS-indexed for grounding.    |

## 7.4 Baselines and evaluation

- **Baselines:** (a) raw SLM, no mitigation; (b) chain-of-thought prompting (BenHalluEval’s tested fix); (c) uniform RAG (retrieve-then-answer, no triage); (d) uniform self-verification (chain-of-verification). SETU must beat all four.
- **Primary metric:** dual-track BenHalluScore (lower is better) on each of the four tasks — penalizes both false alarms and missed hallucinations.
- **Secondary metrics:** task accuracy / F1, abstention rate and over-abstention rate, retrieval recall@k, and calibration (ECE / AUROC) for RQ1.
- **Statistics:** multiple seeds with paired significance tests between SETU and each baseline — the same rigorous protocol the team used in its federated-learning thesis.

## 7.5 Ablations (this is what proves method-level novelty)

- Remove claim decomposition (verify whole answer at once).
- Remove triage (apply one uniform fix to every flagged claim).
- Remove cross-lingual fallback (Bengali retrieval only).
- Remove belief-consistency (retrieval for everything).
- Remove abstention (always answer).

Each removed component should measurably raise BenHalluScore, demonstrating the gain lives in the routing mechanism — not in dataset choice.

## 7.6 Compute feasibility (Kaggle)

- **All inference, no pre-training.** 1.5–3B SLMs run at 4-bit on a Kaggle P100/T4; LoRA is available if any light fine-tuning of the triage classifier is needed.
- **Uncertainty cost is bounded:** semantic entropy needs ~5–10 samples per flagged claim plus a small NLI model — tractable because it runs only on flagged claims, not every token.
- **Retrieval:** BGE-M3 / multilingual-e5 embeddings + FAISS over a chunked Wikipedia subset; no API cost.

## 8. Timeline (approx. 6 months)

| Month | Milestone                                                                                                                                     |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Read Group A + B; finalize proposal; supervisor sign-off; download datasets and Wikipedia; build FAISS index; set up SLM inference on Kaggle. |
| 2     | Claim decomposition + uncertainty scoring; run the calibration study (RQ1) on native and code-mixed input.                                    |
| 3     | Build the triage router (RQ2); implement data-driven retrieval fix with cross-lingual fallback.                                               |
| 4     | Implement reasoning-driven belief-consistency fix + abstention; integrate full SETU pipeline; first end-to-end BenHalluScore run.             |
| 5     | Baselines, ablations (RQ3), statistical tests, error analysis, over-abstention study (RQ4); freeze results.                                   |
| 6     | Thesis writing (Chapters 1–5), presentation, defense; optional BLP / LoResLM workshop submission.                                             |

## 9. Risks and Mitigations

- **Uncertainty signals may be poorly calibrated in Bengali.** This is not a failure — it is RQ1’s finding and a publishable result. If all signals are weak, use retrieval-answerability as the primary flag.
- **Sparse Bengali retrieval corpus.** The cross-lingual English fallback is designed exactly for this; report how often it triggers.
- **Triage misclassification.** Report triage accuracy separately; a claim wrongly routed still gets a fix, so errors degrade gracefully rather than catastrophically.
- **Compute pressure from multi-sample uncertainty.** Sampling runs only on flagged claims; cap samples at 5–10 and cache NLI results.
- **Citation reliability.** Every reference is pulled from its original venue page; the discredited topic PDF is not used.

## 10. Expected Contributions

1. The first hallucination-mitigation framework for Bengali small language models (native + code-mixed).
2. A novel uncertainty-triggered, type-aware triage that routes data-driven vs reasoning-driven hallucinations to different repairs.
3. The first calibration study of uncertainty signals for Bengali SLMs — a dataset-independent scientific finding.
4. An adaptive cross-lingual retrieval fallback tuned for the sparse Bengali corpus.
5. Uncertainty-grounded abstention for Bengali generation, with an over-abstention analysis.
6. A reproducible mitigation evaluation under dual-track BenHalluScore across four tasks, with full ablations.

## 11. Immediate Next Steps

1. Download and read BenHalluEval and HalluGuard in full; confirm the BenHalluScore formula and dual-track setup.
2. Verify all references on Google Scholar / venue pages and build the BibTeX file.
3. Present Table 1 (positioning) and Section 2 (novelty) plus the SETU pipeline to the supervisor for approval.
4. Acquire datasets (TyDiQA-GoldP Bengali, BanglaCHQ-Summ, SOMADHAN) and build the Wikipedia FAISS index.
5. Stand up the Kaggle SLM inference notebook and run the Month-2 uncertainty calibration study first — it de-risks the whole project early.